NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE | NTU Academy for Professional and Continuing Education

(SCTP) Advanced Professional Certificate

Data Science and AI

2.6 Data Pipelines

Module Overview

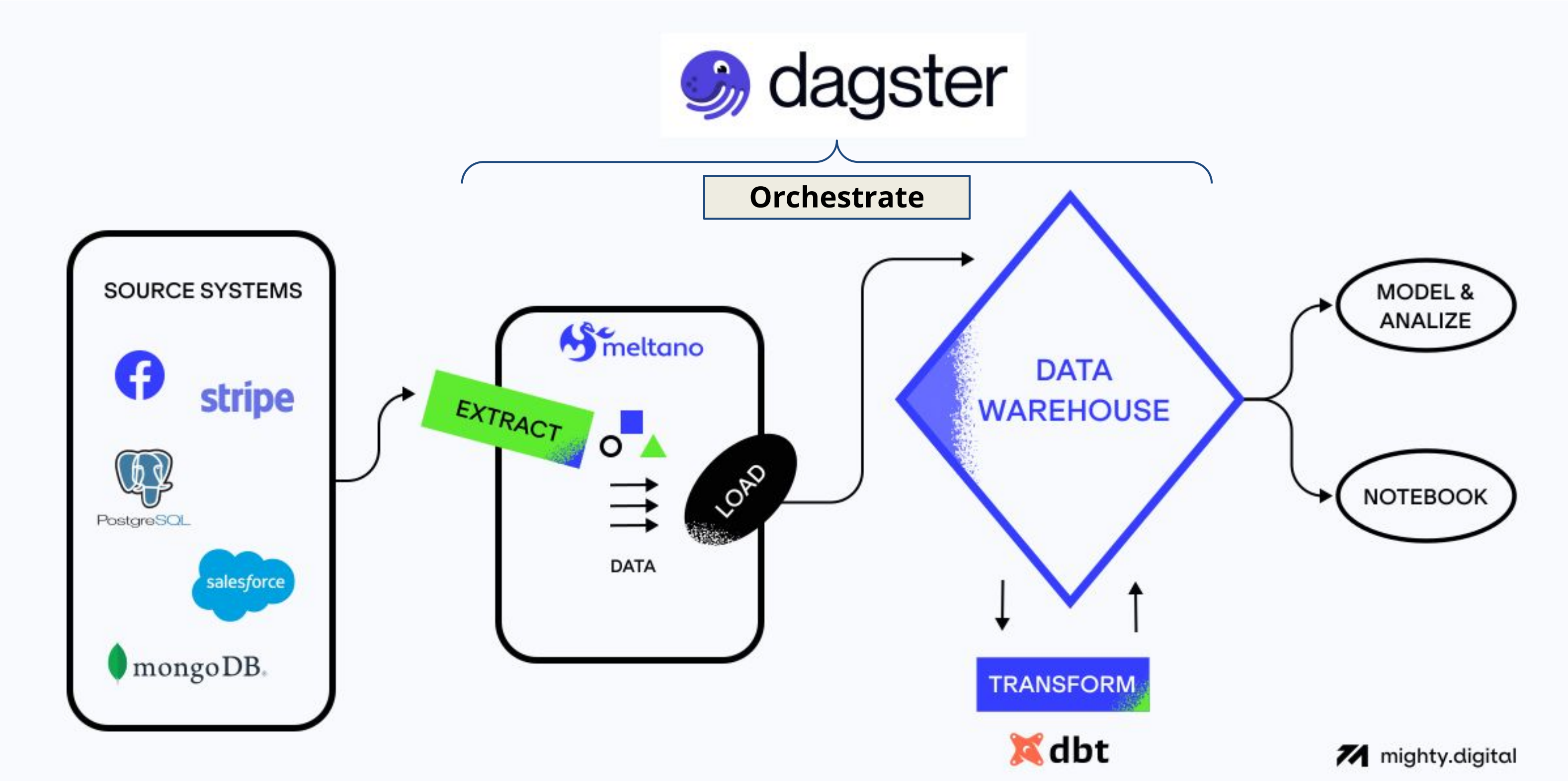
Recap - Dimension Modelling | Surrogate Keys

Type 2 Slowly Changing Dimension

Product Dim (Source)		
Product Name	**Product ID**	**Product Descr**
12 inch box	012	12 inch glued box
10 inch box	010	10 inch ~~glued~~ box 10 inch **pasted** box

Product Dim (Target)					
SID	**Source Product ID**	**Product Name**	**Product Descr**	**EFF_ START_DT**	**EFF_ END_DT**
0001	012	12 inch box	12 inch glued box	Jan-01-1753	Dec-31-9999
0002	010	10 inch box	10 inch glued box	Jan-01-1753	May-12-06
0003	**010**	**10 inch box**	**10 inch pasted box**	**May-12-06**	**Dec-31-9999**

Recap, Self-Study and Prework



Source : https://www.mighty.digital/blog/meltano-the-framework-for-next-generation-data-pipelines

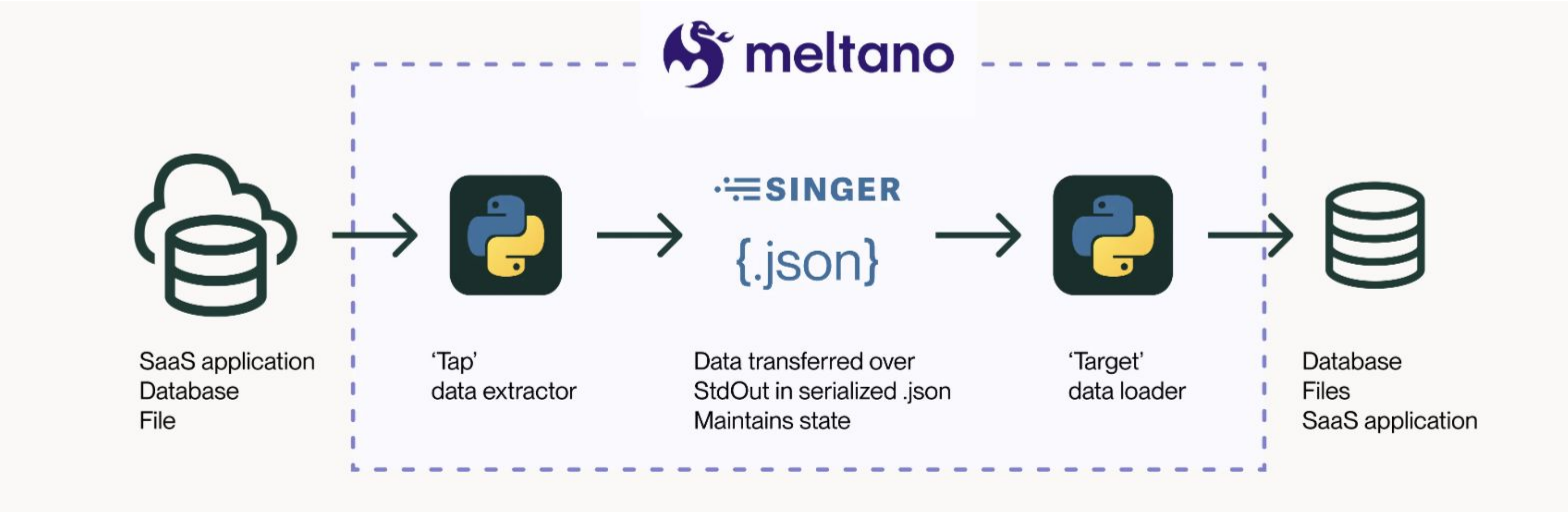
Lesson Objectives

- Create an ELT data pipeline with Meltano and Dbt.
- Orchestrate and schedule a data pipeline with Dagster.

Lesson Plan

1. Part 1: Hands-on with ELT
2. Part 2: Hands-on with Orchestration (Demo)

ELT - Meltano


meltano
SINGER
{.json}
SaaS application
Database
File
'Tap'
data extractor
Data transferred over
StdOut in serialized .json
Maintains state
'Target'
data loader
Database
Files
SaaS application

Orchestration - Dagster

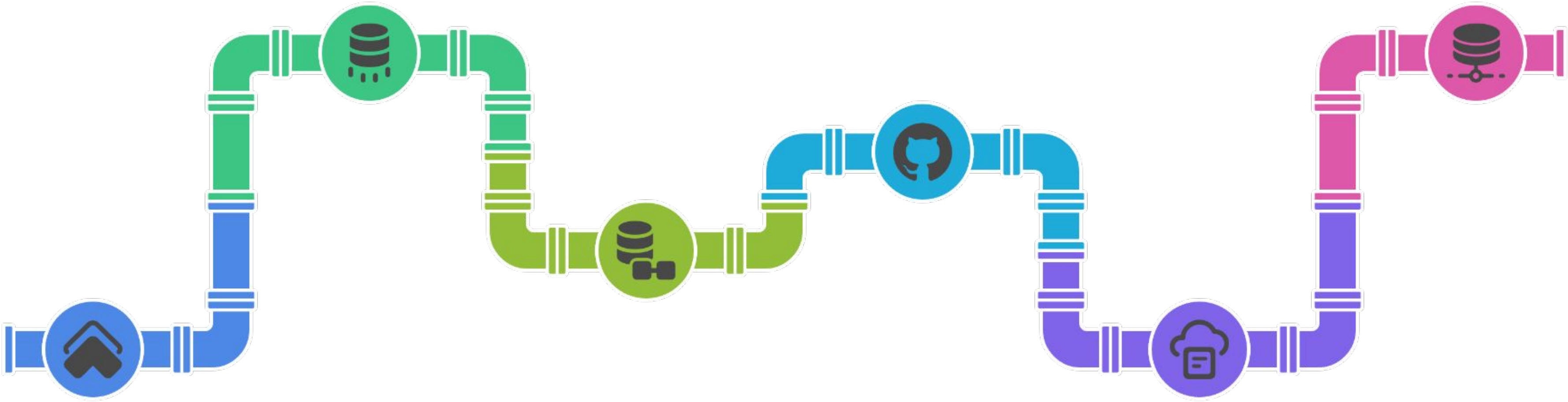
Feature	Dagster	Running Scripts
Automation & Reliability	Built-in orchestration, robust error handling	Manual execution, prone to errors
Observability	Rich UI, structured logs, lineage tracking	Minimal, fragmented logs
Data Quality	Asset-centric, built-in quality checks	Manual, limited or absent
Modern Integrations	Native support for data stack tools and platforms	Requires custom, manual integration

dagster

First-class data assets: Orchestration with context

Dagster lets you orchestrate everything in your data platform—not just dbt. You can run ingestion, transformation, and ML workflows in one place, with unified scheduling and observability. When something fails, you'll know exactly where and why.

Melanto - Pipeline exercise



01
Create Meltano Project
Initialize a new Meltano project

02
Add GitHub Extractor
Configure an extractor to fetch data from GitHub

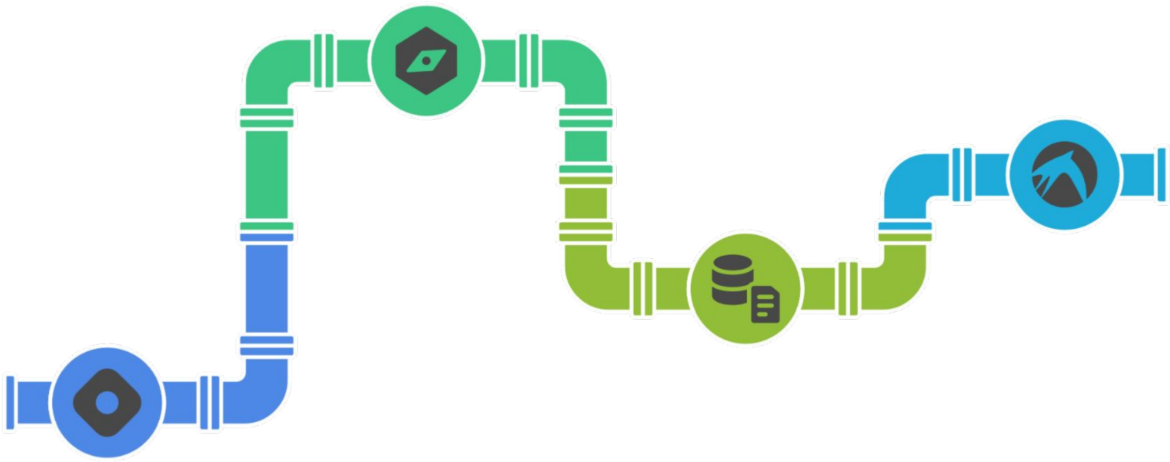
03
Add Dummy Loader
Set up a loader to output data to JSON format

04
Test GitHub to JSON
Run a test to ensure data flows correctly from GitHub to JSON

05
Add BigQuery Loader
Configure a loader to load data into BigQuery

06
Add Postgres Extractor
Configure an extractor to fetch data from Postgres

Dbt Project Setup and Execution



01

Activate dwh environment

Activate the data warehouse environment using conda

02

Create new Dbt Project

Initialize a new Dbt project named resale_flat

03

Create source and models

Define data sources and models within the project

04

Run Dbt

Execute the Dbt project to transform data

Dagster Orchestration Part I

Setup Environment

Create and activate a conda environment for Dagster

Create Dagster Project

Scaffold a new Dagster project and navigate to its directory

Configure GitHub Access

Set up environment variables for GitHub API authentication

Create Data Assets

Define data extraction and transformation assets for pandas releases

Define Pipeline Components

Configure job definitions, scheduling, and pipeline dependencies

Run and Monitor Pipeline

Launch Dagster web interface and execute pipeline through UI

Code-along

Activity

Vscode


BREAK
TIME!!!

End of Lesson - Exit Ticket

Survey Link

https://www.menti.com